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1. INTRODUCTION

Climate change has become one of the most serious global issues, causing significant environmental, social, and economic impacts. Natural disasters such as floods, cyclones, droughts, heatwaves, and storms lead to enormous economic losses every year. Predicting these losses is essential for governments, policymakers, insurance companies, and disaster management authorities to improve preparedness and reduce financial risks.

This project, **"Climate Risk and Economic Losses Using Machine Learning,"** aims to predict economic losses caused by climate-related disasters by analyzing historical climate risk data. Machine Learning algorithms are employed to identify patterns and relationships between climate indicators and economic damages. The developed prediction model assists in estimating future economic losses, thereby supporting effective disaster planning and sustainable development.

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2. OBJECTIVES

The primary objectives of this project are:

- To analyze climate risk data and identify important factors affecting economic losses.
- To preprocess the dataset by handling missing values and selecting relevant features.
- To perform Exploratory Data Analysis (EDA) for understanding the data distribution.

- To implement multiple Machine Learning regression algorithms.
- To compare the performance of different regression models.
- To develop an efficient prediction system for estimating climate-related economic losses.

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3. DATASET OVERVIEW

The project utilizes the **Climate Risk Index (CRI) Dataset**, which contains historical information about climate-related disasters across different countries. The dataset includes several indicators such as:

- Climate Risk Rank
- Climate Risk Score
- Number of Fatalities
- Fatalities per 100,000 Population
- Economic Losses (% of GDP)
- Country Information
- Disaster Statistics

These attributes provide valuable insights into the relationship between climate events and their economic consequences.

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4. DATA PREPROCESSING

Data preprocessing is an essential step before building Machine Learning models. The following preprocessing techniques were applied:

- Imported the dataset into Python.
- Identified and handled missing values using Median Imputation.
- Removed unnecessary attributes.
- Selected relevant features for prediction.
- Split the dataset into training and testing sets.
- Prepared clean and normalized data for model training.

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5. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was performed to understand the characteristics of the dataset. Various statistical summaries and visualizations were used to identify data patterns, feature relationships, and possible outliers. Correlation analysis helped determine which climate indicators have the strongest influence on economic losses.

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6. MACHINE LEARNING MODELS

The following regression algorithms were implemented and evaluated:

- Linear Regression
- Support Vector Regression (SVR)
- Decision Tree Regression
- Random Forest Regression

Each model was trained using the processed dataset and tested to determine its prediction accuracy.

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7. EVALUATION METRICS

The performance of the regression models was evaluated using:

- R^2 Score
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

These evaluation metrics helped compare the effectiveness of each algorithm and identify the most accurate model.

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8. RESULTS

The implemented Machine Learning models successfully predicted climate-related economic losses using historical climate data. Among the evaluated algorithms, **Random Forest Regression** produced the best overall performance due to its ability to model complex relationships and reduce overfitting. The developed prediction system demonstrates the usefulness of Machine Learning techniques for climate risk assessment and economic loss estimation.

9. TOOLS AND TECHNOLOGIES

The project was developed using the following technologies:

- PYTHON (GOOGLE COLAB NOTEBOOK).
- LIBRARIES USED: Pandas, NumPy, Scikit-learn, Matplotlib

10. CONCLUSION

The project demonstrates that Machine Learning can effectively predict economic losses resulting from climate-related disasters. By analyzing historical climate risk data and implementing multiple regression algorithms, the system provides reliable predictions that can support disaster preparedness, financial planning, and policy formulation. The project highlights the importance of data-driven approaches in addressing climate change challenges.

11. FUTURE SCOPE

Future enhancements of the project include:

- Integration of real-time climate datasets.
- Implementation of advanced algorithms such as XGBoost, LightGBM, and Deep Learning models.
- Incorporation of GIS and satellite imagery for improved prediction accuracy.
- Development of a cloud-based web application for real-time climate risk prediction.
- Expansion of the dataset to include additional environmental and socio-economic factors.